

Amulet Motion Controller Fabrication Document

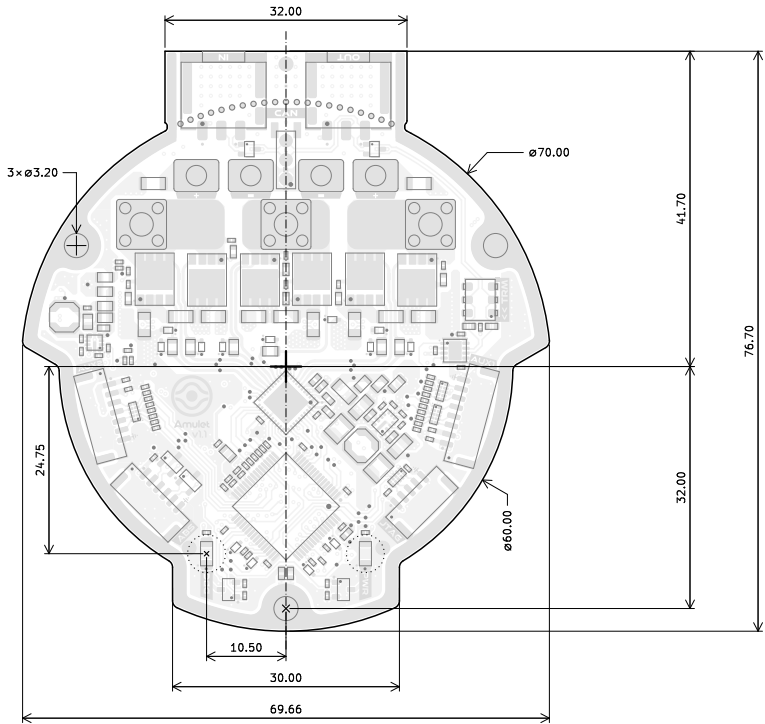
Top Fabrication (Scale 1:1)

Layer Stack Legend

	Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste				Paste Mask	
	F.Silkscreen				Legend	GBR
	F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	Copper	L1 (Sig, PWR)	0.07mm (2oz)		Signal	GBR
	Prepreg		0.18mm	FR4_7628	Dielectric	
	Copper	L2 (GND)	0.035mm (1oz)		Plane	GBR
	Core		0.4mm	FR4	Dielectric	
	Copper	L3 (Sig, PWR)	0.035mm (1oz)		Signal	GBR
	Prepreg		0.18mm	FR4_7628	Dielectric	
	Copper	L4 (Sig, PWR)	0.035mm (1oz)		Signal	GBR
	Core		0.4mm	FR4	Dielectric	
	Copper	L5 (GND)	0.035mm (1oz)		Plane	GBR
	Prepreg		0.18mm	FR4_7628	Dielectric	
	Copper	L6 (Sig, PWR)	0.07mm (2oz)		Signal	GBR
	B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen			Direct Printing	Legend	GBR
	B.Paste				Paste Mask	

Total thickness: 1.66mm

Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX. DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DWG FOR REF ONLY.
- 2) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS. SELECTED HOLE LOCATIONS SHOWN ON THIS DWG FOR REF ONLY.
- 3) IMPEDANCE CONTROL REQUIRED. Microstrip 100-Ohm Differential (L1 ref. L2) 0.2032mm width, 0.28mm spacing
- 4) CONFIRM TRACE WIDTHS AND SPACINGS.
- 5) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

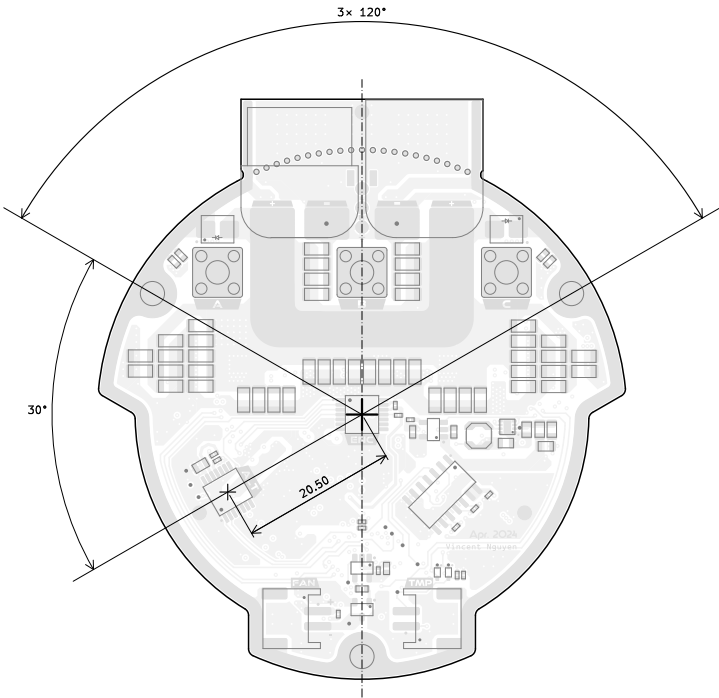
TRACE WIDTH	0.20 mm
TRACE TO TRACE	0.20 mm
MIN. HOLE (PTH)	0.25 mm
MIN. HOLE (NPTH)	0.70 mm
ANNULAR RING	0.15 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.25 mm
HOLE TO HOLE	0.254 mm

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: EPFL Xplore Research ABC		Variant: NO_SNUBBER	
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Top Fabrication	File Name: amulet_controller.kicad_pcb	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
	Sheet Path:	Reviewer:		Size: A4	Sheet: 1 of 11

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Bottom Fabrication (Scale 1:1)



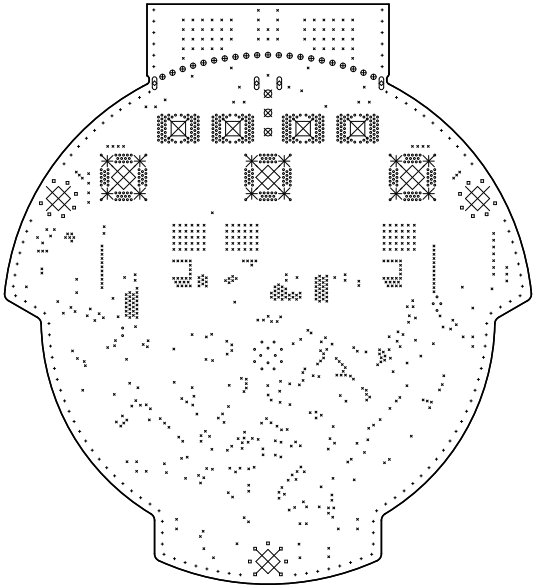
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			Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Bottom Fabrication		File Name: amulet_controller.kicad_pcb	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
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Drill Drawing (Top View) (Scale 1:1)

Symbol	Count	Hole Size	Plated
•	674	0.25mm	Plated
•	438	0.25mm	Plated
•	120	0.30mm	Plated
▪	21	0.50mm	Plated
◊	4 (slot)	0.60mm	Plated
⊠	3	1.00mm	Plated
✱	12	1.50mm	Plated
⊠	4	2.70mm	Plated
⊠	6	3.20mm	Plated
•	21	0.70mm	Unplated

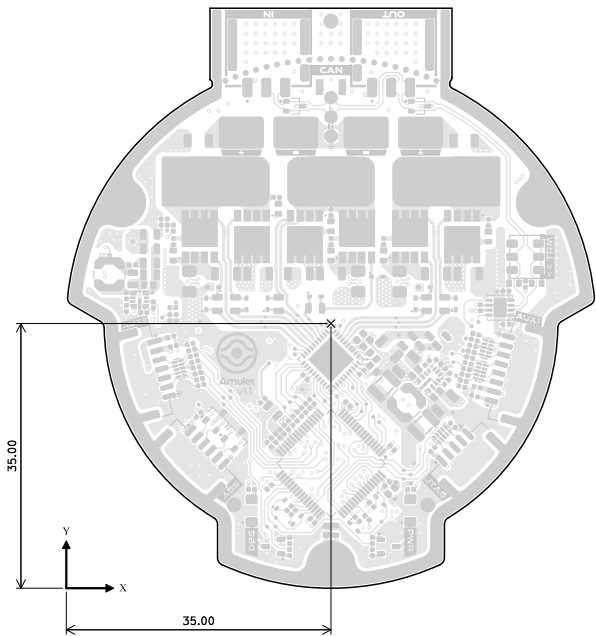


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	Sheet Title: Drill Map	File Name: amulet_controller.kicad_pcb	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
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Ref.	Net	X	Y
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Top Test Points (Scale 1:1)



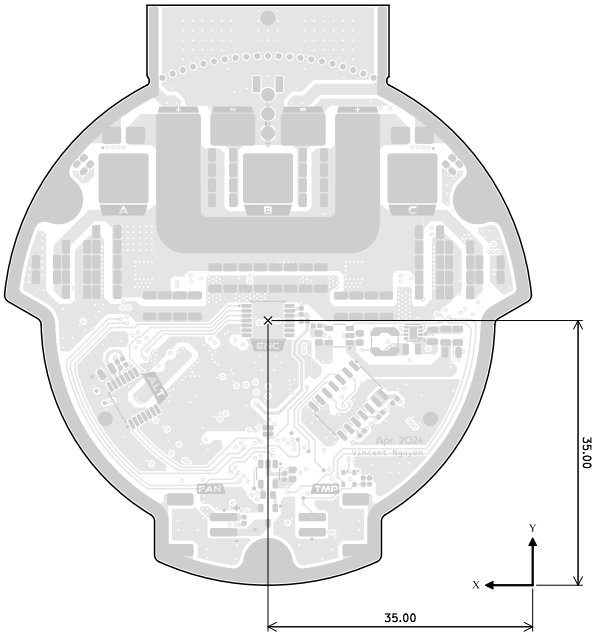
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	Sheet Title: Top Test Points	File Name: amulet_controller.kicad_pcb	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
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Ref.	Net	X	Y
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Bottom Test Points (Scale 1:1)

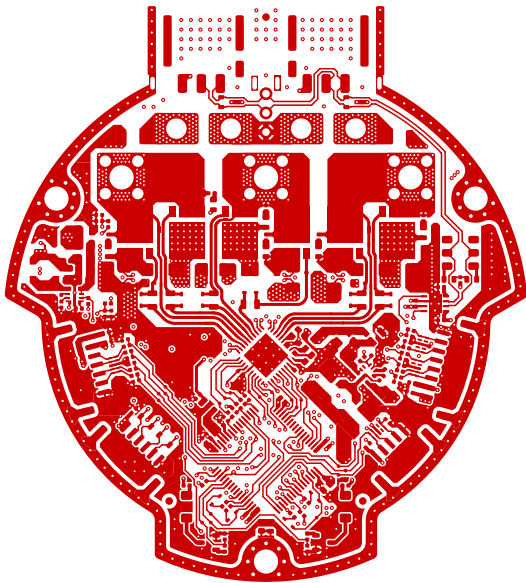


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	Sheet Title: Bottom Test Points	File Name: amulet_controller.kicad_pcb	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
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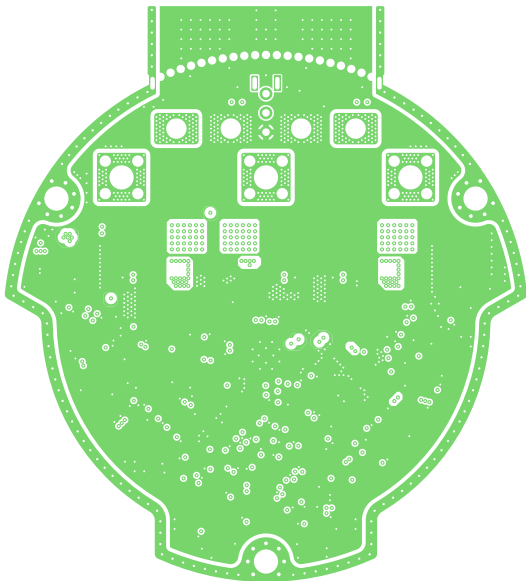
L1 (SIG, PWR)



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	Sheet Title: F.Cu	File Name: amulet_controller.kicad_pcb	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1
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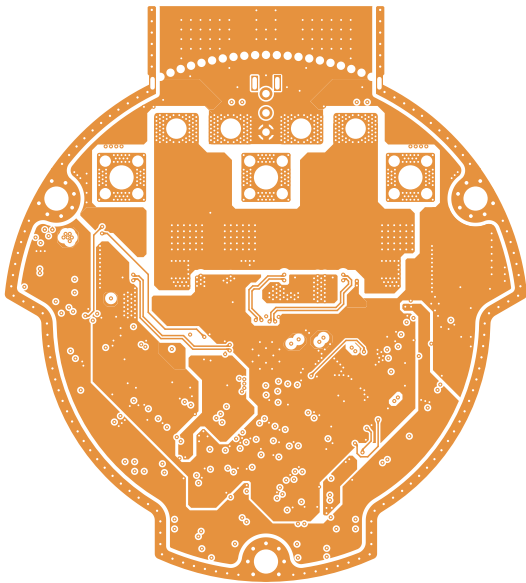
L2 (GND)



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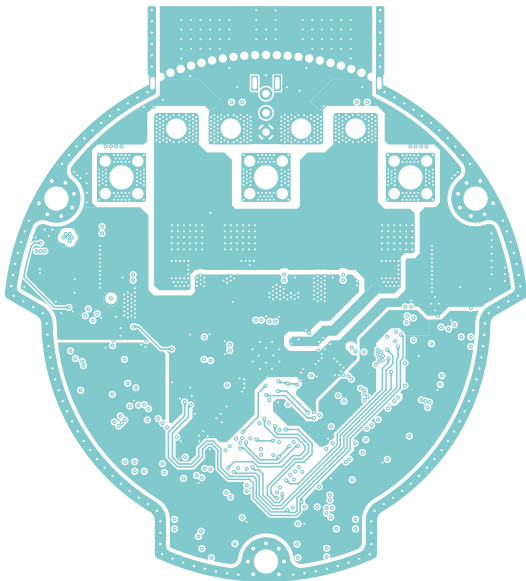
L3 (SIG, PWR)



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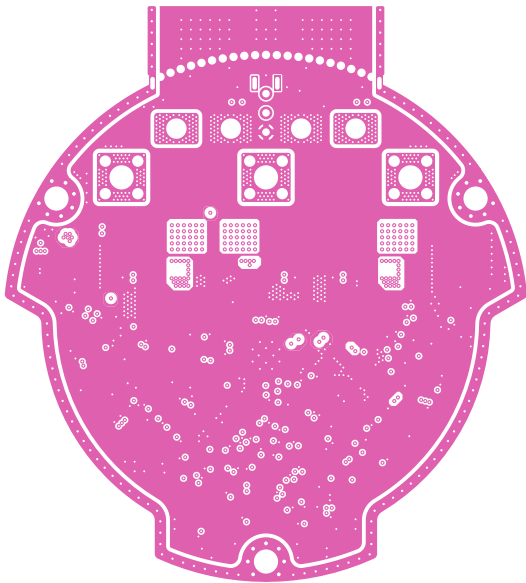
L4 (SIG, PWR)



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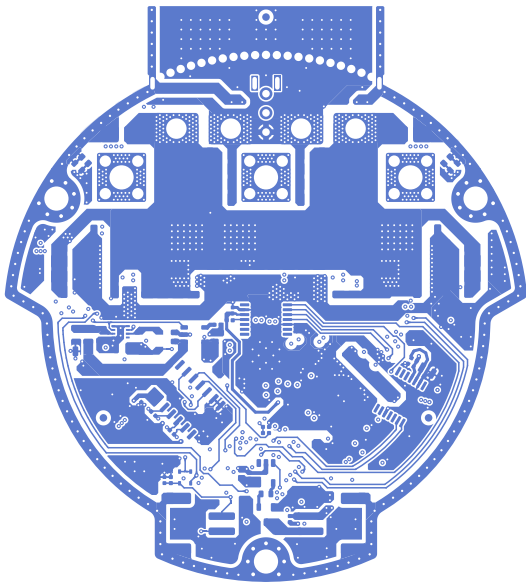
L5 (GND)



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L6 (SIG, PWR)



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	Sheet Title: B.Cu	File Name: amulet_controller.kicad_pcb	Designer: Vincent Nguyen		Date: 2024-04-13		Revision: 1.1
	Sheet Path:		Reviewer:		Size: A4	Sheet: 11 of 11	